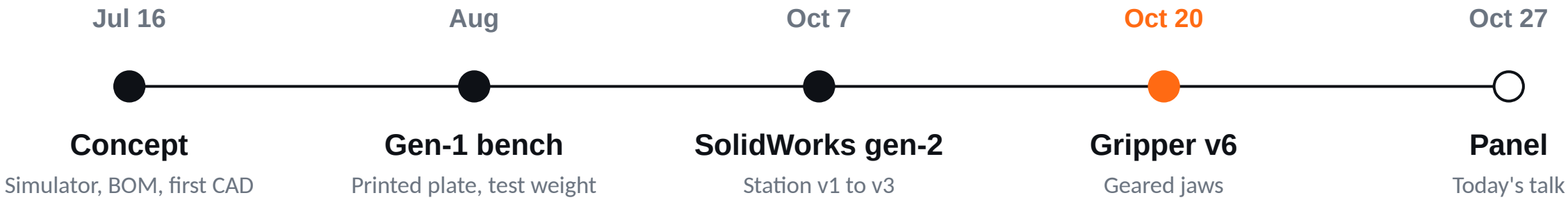
RoomCleaner

A four-cable ceiling robot
designed to pick laundry off the floor

Jasper Buntinx · Technical presentation · Oct 27, 2026



Solo project: a four-winch cable robot, a camera claw, and the control architecture.



BOUGHT

Stepper motors, A4988 drivers, ESP32 boards, DROK buck, MG996R servo, bearings, pulleys, line.

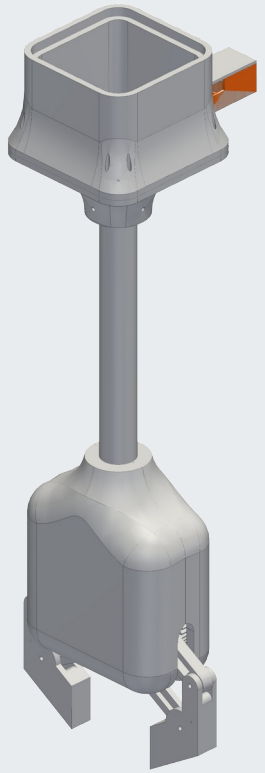
MADE BY ME

Station and claw design in SolidWorks, printed and machined parts, bench wiring, test plan.

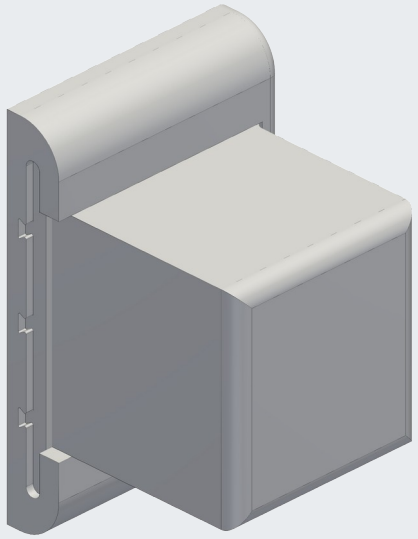
HOW TO READ THE TAGS

MEASURED	On hardware, with an instrument
CALCULATED	Hand or CAD calculation
SIMULATED	Run in a model or simulator
PLANNED	Not done yet

Done means jeans from anywhere on the floor into the hamper.



Claw v6 with camera bay

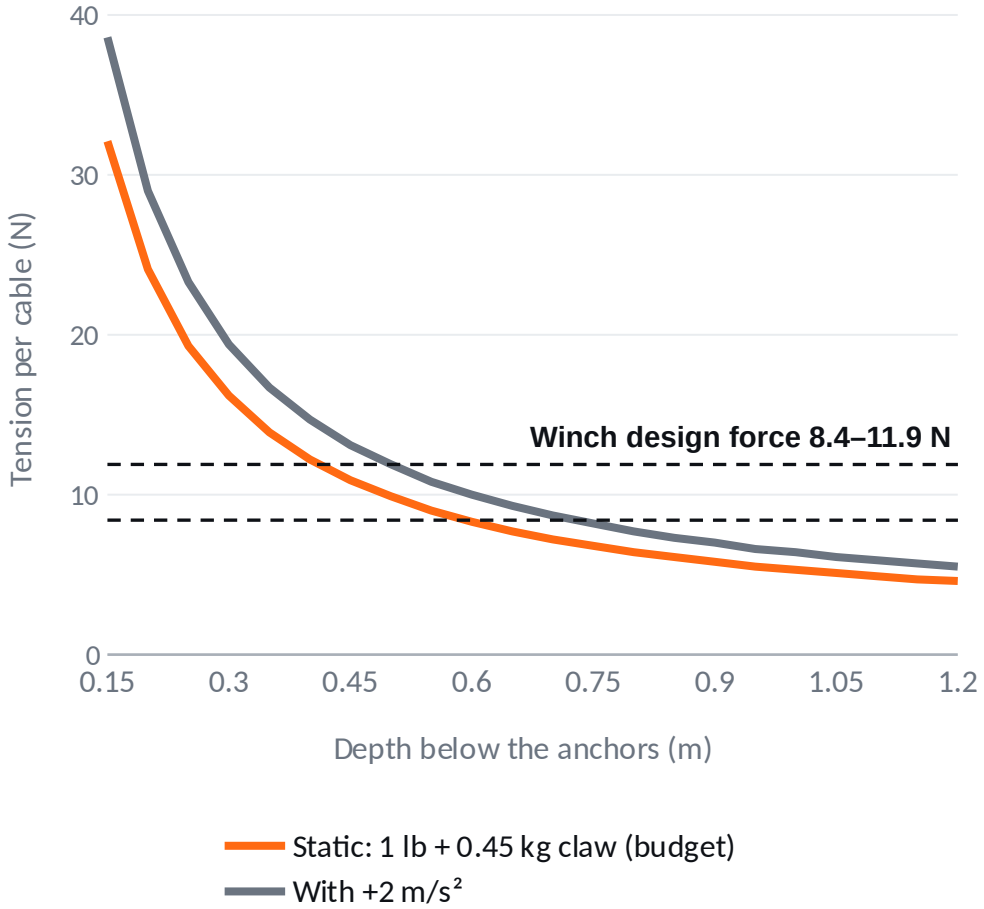


Winch station v3

0.9 kg	Provisional jeans payload plus 2 m/s²; first gate 1 lb
≤ 450 g	Claw mass budget
4 × studs	Stations designed to screw into wall studs
1 switch	The switched power strip is the only kill switch
0 people	Nobody under the robot while it moves

CALCULATED

Cable angle sets the limits: walls bind at task height, ceiling above.



1	WANTED	Use the whole room.
2	REALIZED	**32 N at 0.15 m depth, 8.3 N at 0.6 m.**
3	DECIDED	Ceiling margin 0.4 m, plus a wall margin.
4	RESULT	Max 0.5 kg at 0.4 m, 1.1 kg at task height.
5	BUT	Claw mass counts like payload; 0.45 kg is a budget.

FUNDAMENTALS

$T = W / (4 \sin \varphi)$ (claw centred, equal share)

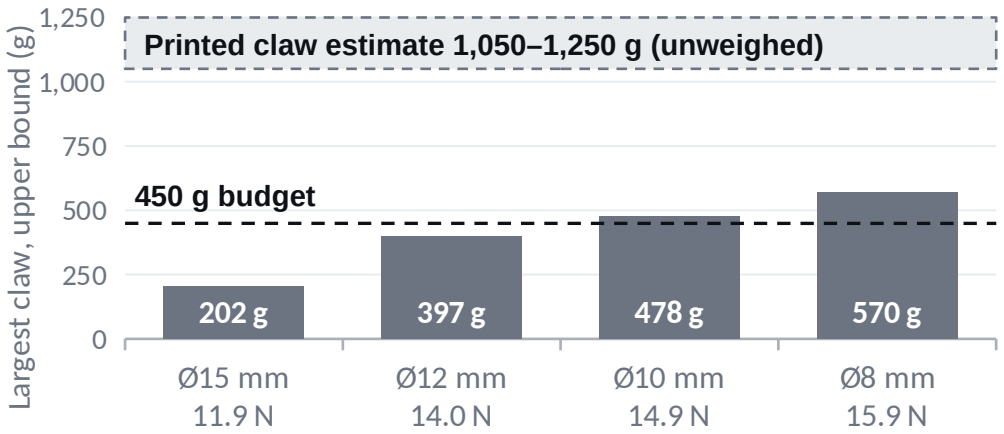
$\sin \varphi = d / \sqrt{(d^2 + D_h^2)}$, $D_h = 2.17$ m, $W = 8.86$ N: $T = 32.1$ N at 0.15 m, 8.3 N at 0.60 m. Off-centre needs a linear program.

MEASURED
CALCULATED

The driver, not the motor, capped my pull; the fix is open.

0.79–0.85 V → 0.99–1.06 A

Vref pot maximum (two readings) and the current it allows



Spool core and one-phase design force

PLANNED — not yet run

Luggage-scale pull at 1.06 A, empty and full spool; settles the 0.7 factor and one- vs two-phase torque.

1	WANTED	Raise current from 1.0 to 1.35 A.
2	REALIZED	**Vref pot maxes at 0.85 V: 1.06 A cap.**
3	DECIDED	Run at the cap; open: core +25 %, or 3:1 gearing.
4	RESULT	Winches hold ≤ ~1.1 kg; claw plus jeans ~2 kg.
5	BUT	Full-spool pull 12.6–17.9 N; pull test decides.

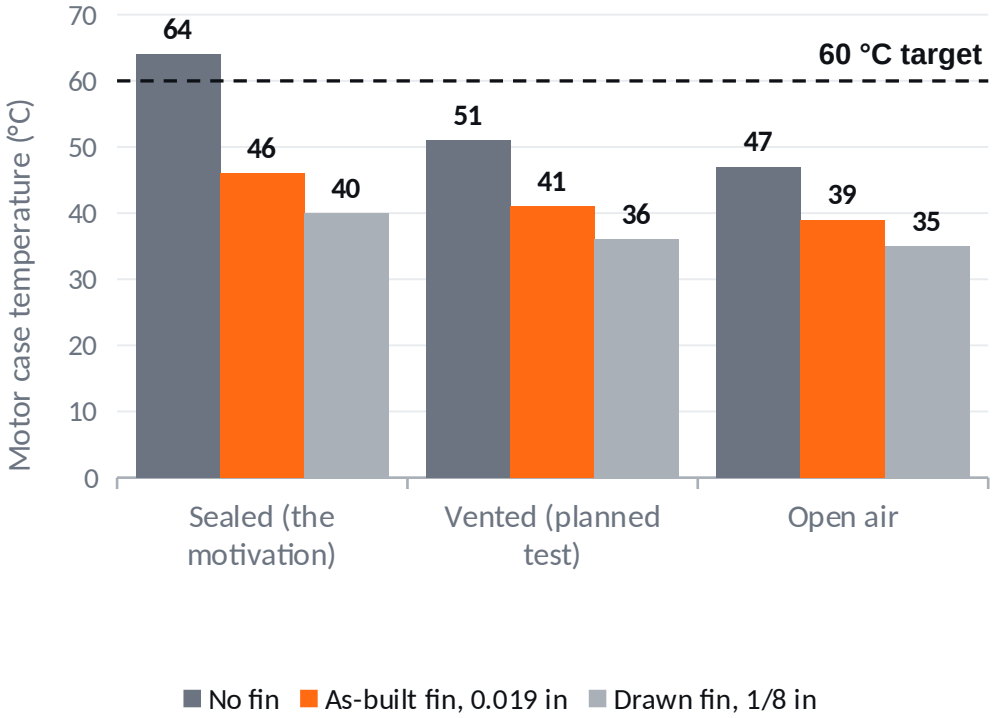
FUNDAMENTALS

$$I = V_{ref} / (8 \cdot R_s) = 0.85\ V / 0.8\ \Omega = 1.06\ A$$

$\tau_{run} = 0.7\ \tau_{hold}$, with $\tau_{hold} = 0.255$ N·m (one-phase) or 0.18 N·m (two-phase). $T = \tau / r$: at r = 10 mm 17.9 or 12.6 N; at the core 23.8 or 16.8 N.

CALCULATED

A sealed station would heat the motor mount, my most loaded joint.



PLANNED — not yet run

Hold test, vented shell, 1.06 A: model predicts 41 °C (as-built fin); pass below 60 °C.

1 **WANTED** — A clean, closed printed shell.

2 **REALIZED** — Holding stepper heats (I^2R); PETG conducts 1/800 of aluminium.

3 **DECIDED** — Vents plus an aluminium fin.

4 **RESULT** — **Sealed 64 to 46 °C with fin; vented 51 to 41 °C.**

5 **BUT** — No thermometer data yet; vented hold test planned.

FUNDAMENTALS

$$P = I^2R;\quad R_{cond} = L / (k\,A);\quad R_{conv} = 1 / (h\,A)$$

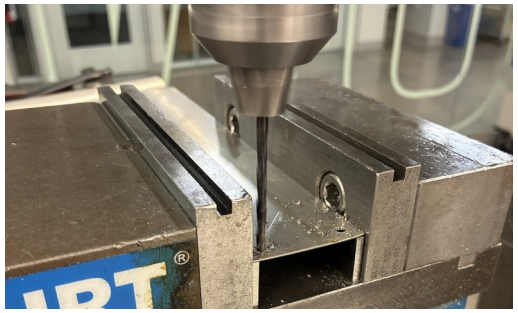
$P = 1.06^2 \times 1.8\ \Omega = 2.0$ W (R assumed). Flange path: 5 mm PETG 14 K/W vs 3.2 mm aluminium 0.011 K/W (1260× = 800× k, 1.6× thickness).

MEASURED
CALCULATED

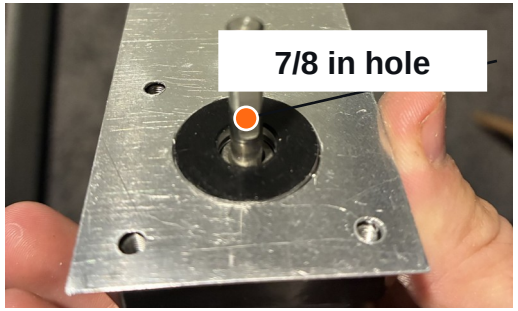
Drawn at 1/8-inch, built at 0.019-inch, then rechecked the physics.



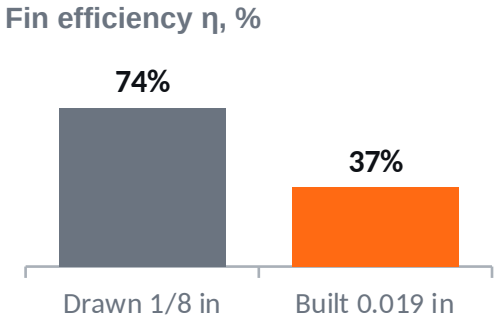
1 Sheared blanks, shop drawing



2 Drilling the holes on the mill



3 Fitted: boss centred, 4 holes line up



PLANNED — not yet run

Hold test: does the thin sheet bow or buzz, and are the screw threads deep enough?

1	WANTED	Drawing: 1/8 in plate, fin efficiency 74 %.
2	REALIZED	As built: 0.019 in (0.48 mm) sheet, not the 1/8 in drawn.
3	DECIDED	**Re-ran the fin equation: 74 % falls to 37 %.**
4	RESULT	Still carries about half the heat: 46 °C.
5	BUT	May buzz; screws reach 2.7 mm deeper.

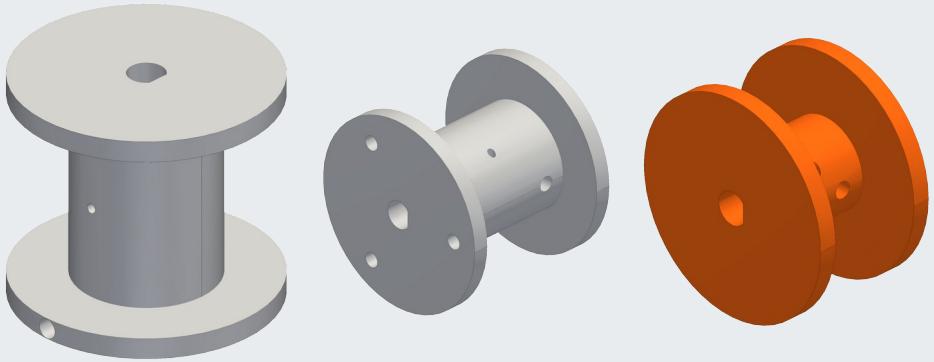
FUNDAMENTALS

$$\eta = \tanh(mL) / (mL), \quad m = \sqrt{(2h / (k\,t))}$$

t = 0.483 mm: m = 13.5 /m, mL = 2.69, η = 37 %, R_{fin} = 22 K/W. Drawn 1/8 in: mL = 1.05, η = 74 %, R_{fin} = 10.9 K/W.

CALCULATED

My first spool outran its motor shaft; the fix changed the math.



Gen-1 spool
Ø20 core, 32 mm long

v2
bore Ø5.2, too tight

v3
Ø15 core, bore Ø5.45

Ø20 → Ø15
core, mm

32 → 22
overall length, mm

Ø5.2 → Ø5.45
bore, mm

1 **WANTED** — 32 mm spool (26 mm drum) on a Ø20 core.

2 **REALIZED** — About 14.5 mm past the 24 mm shaft (CAD).

3 **DECIDED** — **Ø15 core, 16 mm drum; radius tracks wound length.**

4 **RESULT** — v2 bore too tight: Ø5.2 opened to Ø5.45.

5 **BUT** — Printed holes shrink; no caliper reading yet.

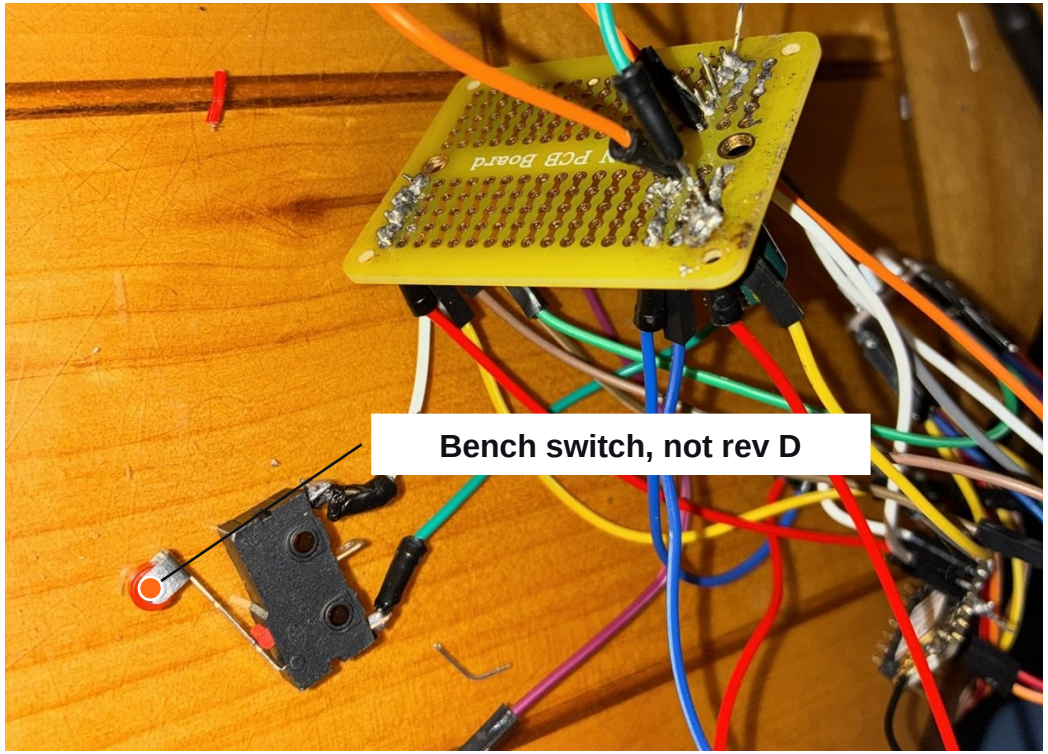
FUNDAMENTALS

$$r(n) = r_0 + n \cdot d \cdot p; \quad T = \tau / r(n); \quad \ell = 2\pi r$$

20 ft of line, diameter assumed 0.5 mm (not measured): 5 layers, r = 7.5 to 10 mm, T = 23.8 to 17.9 N (one-phase), line per rev 47 to 63 mm (+33 %).

CALCULATED · CAD

Bead must not cross the pulley; verification caught rev C.



PLANNED — not yet run

Trip test: 5 mm bead on real line against a printed rev D arm.

1	WANTED	Bead on the line trips a home switch.
2	REALIZED	Mid-span, the bead rides over the pulley.
3	DECIDED	A drop arm under the pulley.
4	RESULT	**Rev C missed in X and Y; rev D overlaps 3.75 mm.**
5	BUT	Never trip-tested on real line yet.

FUNDAMENTALS

$$Y\ \text{overlap} = (r_{bead} + r_{roller}) - Y_{roller}$$

(2.5 + 2.25) − 1.0 = 3.75 mm, need ≥ 2.0 mm. Rev C: 5.0 mm off in X, 10.5 mm in Y. X error ≤ 1.0 mm across 18 to 22 mm pulleys.

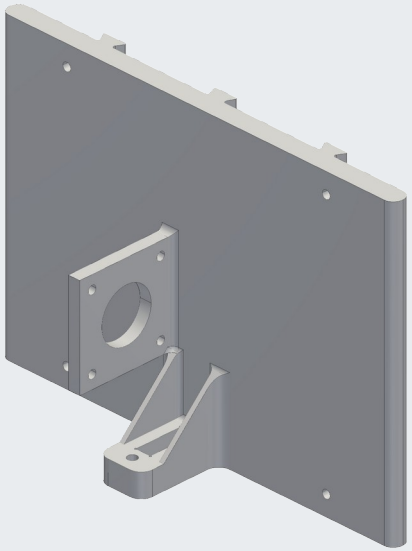
CALCULATED

PLANNED

A removable station moves the anchor from a screw to a slide.



Fixed wall frame



Slide-in housing

PLANNED — not yet run

SolidWorks linear static study (stress, safety factor, deflection) and a station-mount pull test.

1	WANTED	Swap a station without re-calibrating the room.
2	REALIZED	Anchor error maps 1:1; studs see pull-out.
3	DECIDED	Fixed frame plus slide-in; target SF ≥ 3 at 34 N.
4	RESULT	**Hand estimate: tab root 8–14 MPa vs 8.33 limit; study pending.**
5	BUT	Statics can't see creep; pull test too.

FUNDAMENTALS

$$\sigma = M\,c\,/\,I \text{ (rail root)};\quad \delta = F\,L^3\,/\,(3\,E\,I) \text{ (cable exit)}$$

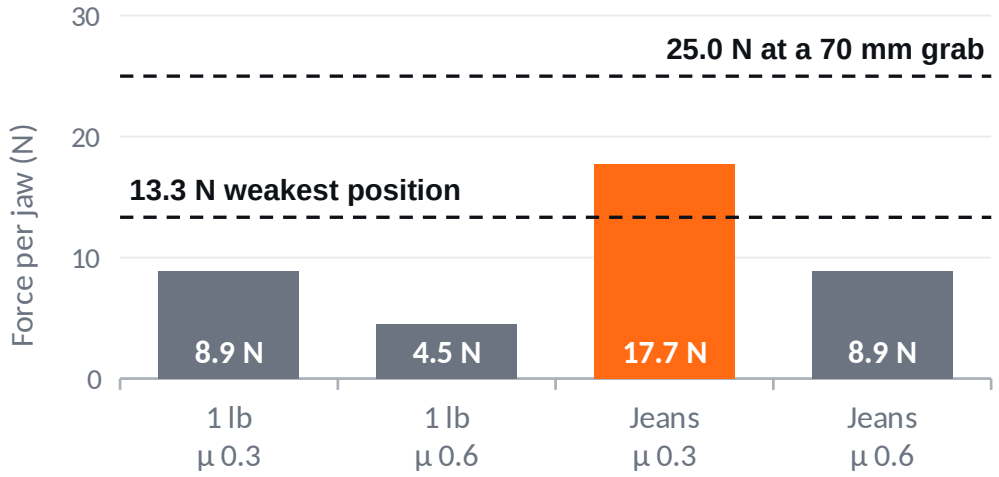
Targets at the 34 N slip load (24 N if two-phase): $\sigma \le 8.33$ MPa (SF 3 on 25 MPa), ≤ 0.5 mm at the cable exit. Direct drive; 3:1 gearing adds ~2.7×.

CALCULATED

SIMULATED

A clamp stalls the servo every grip; teeth are checked at stall.

Bars: needed per jaw · Dashed lines: jaw force available



1 **WANTED** Five soft tendon fingers.

2 **REALIZED** A clamp is simpler but stalls the servo.

3 **DECIDED** Module 1.5, 2:1 ratio, 36T gear rebuilt from my v6 gears.

4 **RESULT** Gears 157 g to 96 g; 257 checks pass with conditions.

5 **BUT** **Jeans at μ 0.3 need 17.7 N; jaws give 13.3–25 N.**



My v6 compound, 3:1

gear set 157 g (CAD)



Rebuilt v6r, 2:1

gear set 96 g (CAD)

FUNDAMENTALS

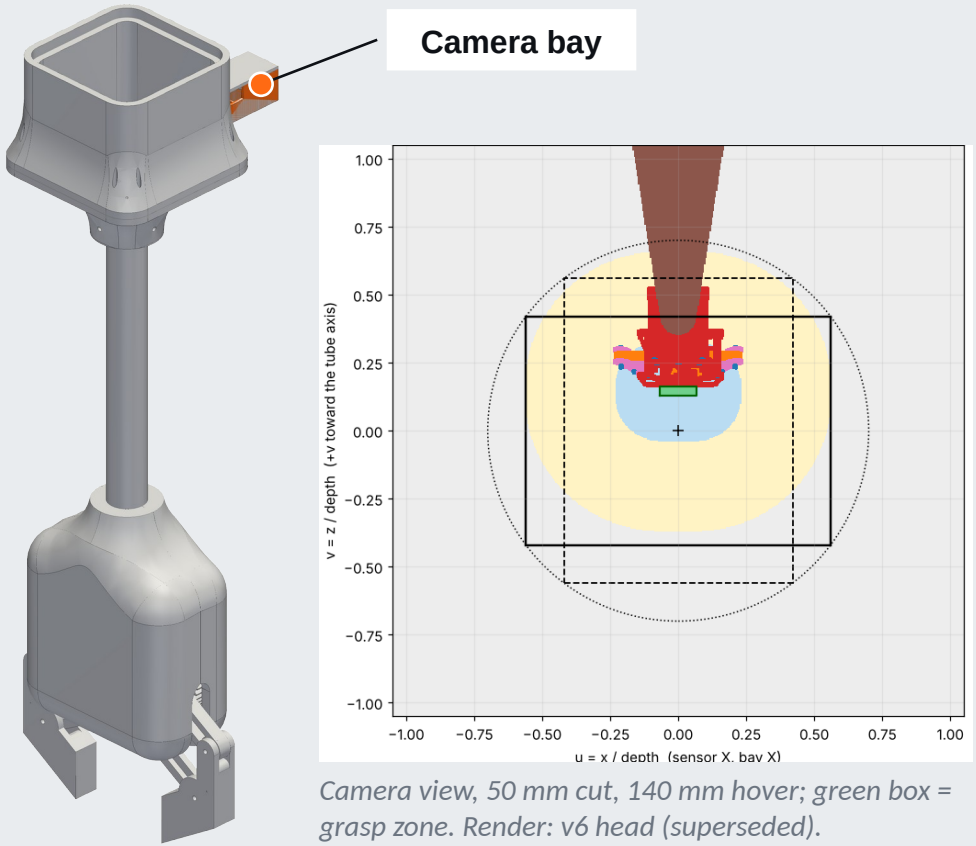
$$\sigma = F_t / (b \cdot m \cdot Y); \quad N \geq m(g + a) / (2\mu)$$

Module 1: SF 0.35–0.41 at stall; module 1.5: teeth 2.0–2.5, 36T web 1.7. Jeans: 0.9 × 11.81 / (2 × 0.3) = 17.7 N per jaw.

CALCULATED

SIMULATED

The camera can't see the grasp at contact, and that's fine.



Camera view, 50 mm cut, 140 mm hover; green box = grasp zone. Render: v6 head (superseded).

1	WANTED	Long aluminium arm; camera sees the jaws.
2	REALIZED	At contact, a plate hides the grasp spot.
3	DECIDED	Cut 50 mm of tube, not 100 mm.
4	RESULT	**Visible from 140 mm hover (was 172); saves ~33 g.**
5	BUT	Last ~14 cm is blind; add a hover correction.

FUNDAMENTALS

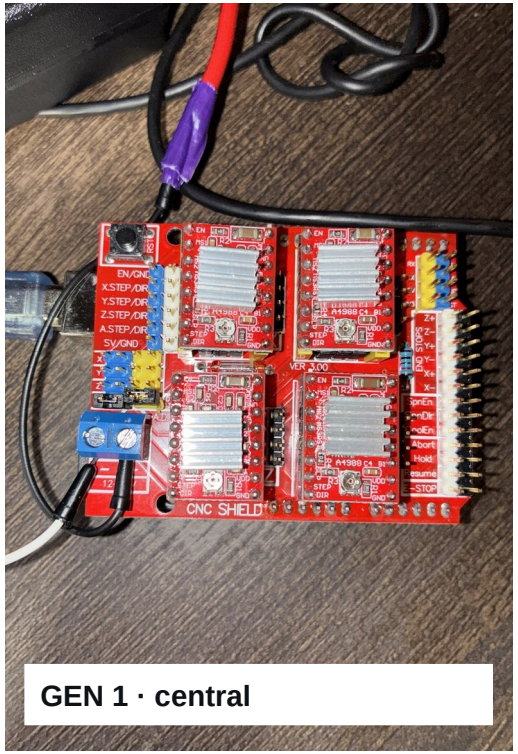
$$D^* = (k\,(H_0 - c) - 93) \,/\, (1 - k), \quad k = 1 - 59.5/97$$

$k = 0.387$, $H_0 = 512$ mm: $c = 0$ gives $D^* = 171$ mm (ray-cast 172), $c = 50$ mm gives 140 mm. Saved 33 g assumes a 3 mm wall.

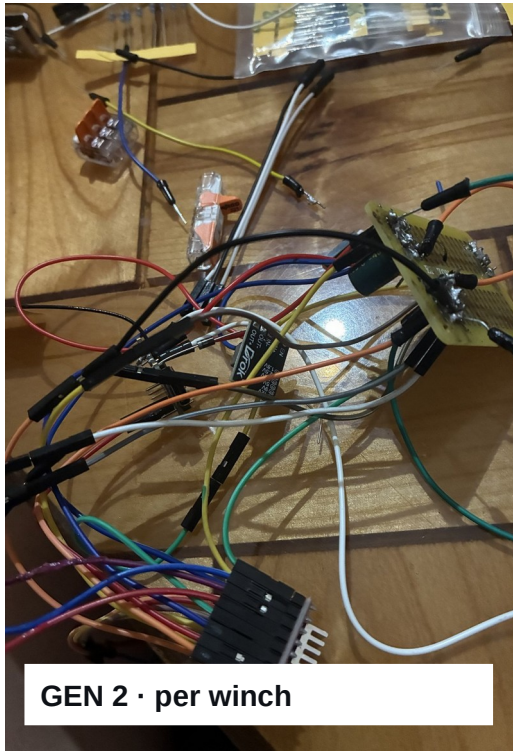
CALCULATED

SIMULATED

One controller per winch removes long harnesses; a review priced sync.



GEN 1 · central



GEN 2 · per winch

PLANNED — not yet run

Four stations, 1 lb, 300 mm at 50 mm/s: pass if each cable is within 2 mm.

1 **WANTED** July: one controller; synchronised motion came free.

2 **REALIZED** 32 ft harnesses; skew becomes force on four cables.

3 **DECIDED** **Host plans; stations run timed segments, own clocks.**

4 **RESULT** Control logic simulated only.

5 **BUT** Not yet run on four stations.

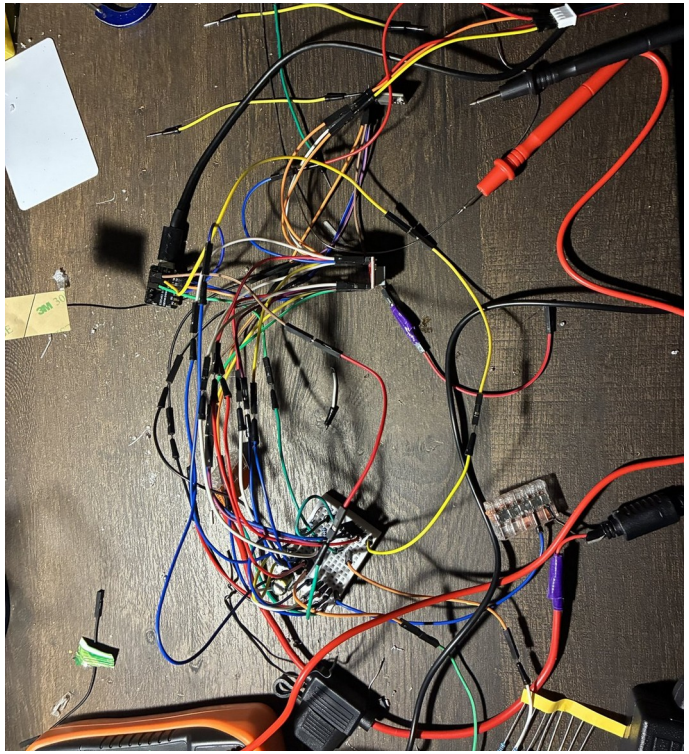
FUNDAMENTALS

$$\Delta\ell = v\cdot\Delta t;\quad F = k\cdot\Delta\ell;\quad V = I\cdot\rho\cdot 2L / A$$

50 mm/s × 50 ms = 2.5 mm; assumed 3 N/mm gives 7.5 N (upper bound). 16 AWG, 32 ft, 2 A (illustrative): 0.51 V drop (18 AWG: 0.82 V).

MEASURED

A motor that wouldn't move taught me to measure before coding.



3.3 V

expected on the logic rail

1.14 V

read on my meter (not logged)

1	WANTED	First motion on the new station board.
2	REALIZED	**Pulses sent, shaft still; logic rail read 1.14 V.**
3	DECIDED	Solder 3.3 V; pull resistors; JP1; measure VDD.
4	RESULT	Motor moved after repair; Rev D layout net-checked, not built.
5	BUT	As-built board still needs its meter checklist.

PLANNED — not yet run

Meter checklist on the as-built board: VDD, SLEEP, ENABLE and the buck output.

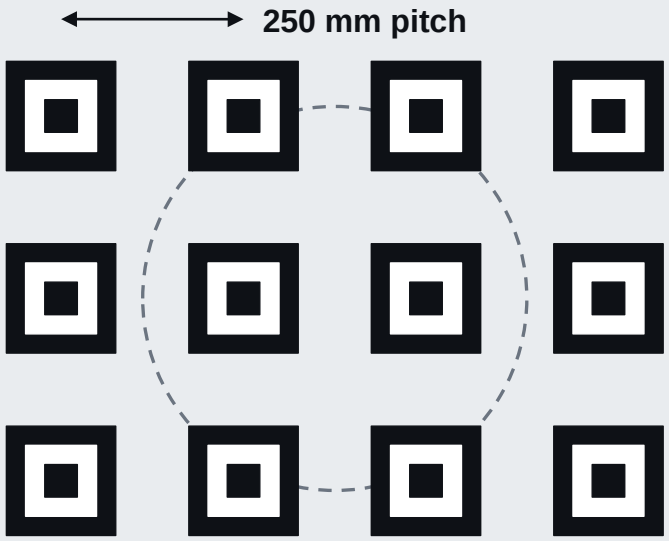
FUNDAMENTALS

$$I = V_{ref} / (8 \cdot R_s) = 0.85\ V / 0.8\ \Omega = 1.06\ A$$

Meter first: logic rail 1.14 V read against 3.3 V expected. The same habit on Oct 21 found the Vref pot topping out at 0.85 V.

CALCULATED
SIMULATED

I stopped buying precision and moved the safety into margins.



2 tags in view: 4.9 mrad; 4 tags: 1.3 mrad

6.6 mm → 0.7 mm

per-frame lever-arm error at 0.51 m: one tag vs four tags in view (two tags: 2.5 mm)

1	WANTED	Absolute cable lengths from vision alone.
2	REALIZED	One tag errs 6.6 mm/frame; cable σ stays above 12 mm.
3	DECIDED	**Margins carry safety; probe slack; drop the IMU.**
4	RESULT	Simulated: 6/6 startups; 3 of 6 reach full trust.
5	BUT	Claw tilts 13–21° near corners; all simulated.

FUNDAMENTALS

$$\delta x = \sigma_\theta \cdot L; \quad \sigma_N = \sigma / \sqrt{N}$$

One tag: 12.9 mrad × 0.51 m = 6.6 mm. Four tags in view: 1.32 mrad × 0.51 m = 0.68 mm; two tags: 4.85 mrad = 2.5 mm.

Measured, calculated, simulated, planned: the test that decides the rest.

MEASURED	CALCULATED	SIMULATED	PLANNED, with pass limits
• Vref pot maximum 0.79–0.85 V • One motor on all four driver channels • Buck output 5.00 V • Logic rail 1.14 V (the fault) **Observed, not instrument-measured** • Fin test-fits on the motor (visual) • 18 V read at the supply vs 12 V label (re-measure)	• Cable statics, 32 to 4.6 N • Gear stress (Lewis) • Fin efficiency, 74 % to 37 % • Wire drop, 0.51 V • Grip force required per jaw	• Control system and sync • Startup calibration • Camera view vs tube length • Jaw force, four-bar model	• Hold test, vented shell: motor case below 60 °C • Winch pull test (luggage scale), empty and full spool • Station-mount pull test • SolidWorks station study: SF ≥ 3 at 34 N, ≤ 0.5 mm • 1 lb, four stations, 300 mm at 50 mm/s: ≤ 2 mm per cable

FIRST TEST

Luggage-scale pull at 1.06 A, empty and full spool; under about 25 N points to a 3:1 reduction (decision open).

Also: confirm VDD = 3.3 V at the driver and measure the line diameter.

WHAT I DELETED, WHAT I LEARNED

My best decisions were deletions.

DELETED

- The IMU
- Five fingers, now two jaws
- 50 mm of tube
- The housing block, now an open frame (−560 g, CAD)
- The 5 lb reserve: a 1 lb gate first

THREE LESSONS

01 **Measure before you trust a margin**

The 0.85 V Vref cap and the 1.14 V rail

02 **Print the coupon before the part**

The spool bore: Ø5.2 became Ø5.45

03 **Stall is the design case**

Gear stall and motor holding heat, not the average

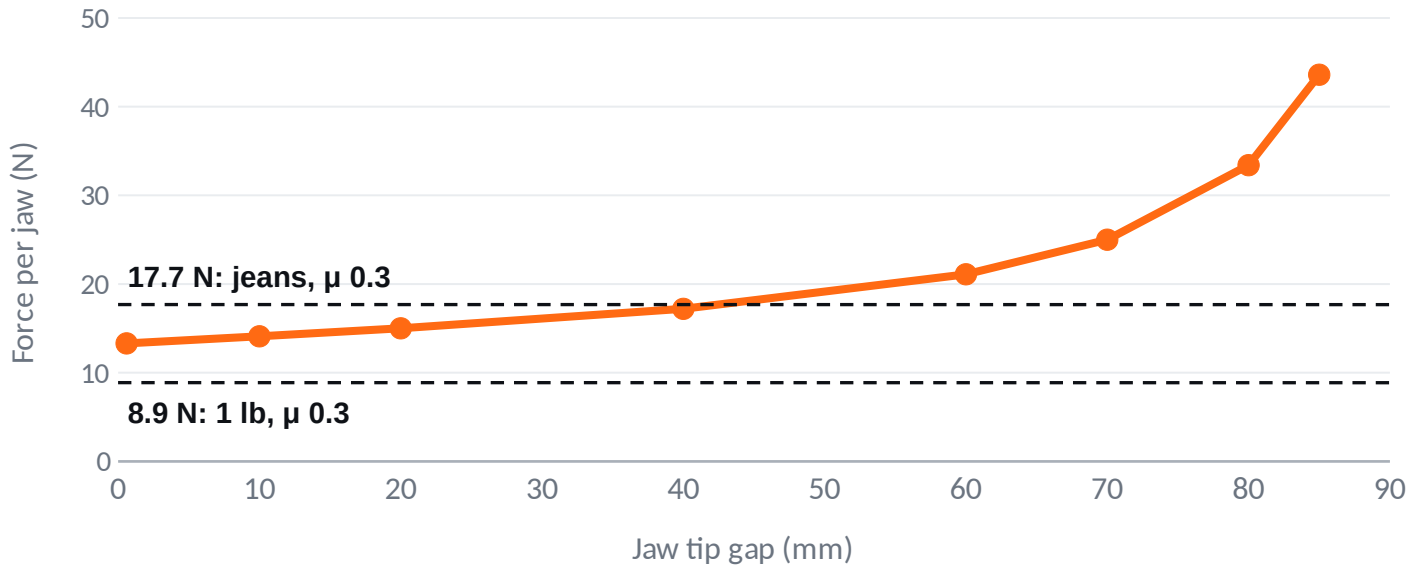
With half the time again: buy pulleys and couplers sooner, and build one station to test before four.

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SIMULATED

CALCULATED

Jaw force is lowest closed and over 3× higher near full open.



76.5°
jaw stroke from a 153° servo (2:1)

91.6 mm
open gap, closed gap 0.000 mm

0.83 N·m
jaw torque at 4.8 V stall

Near full open the four-bar nears toggle: the effective lever arm shrinks (59 mm at a 10 mm gap, 33 mm at 70 mm) and force rises. No toggle inside the stroke.

FUNDAMENTALS

$$F_{jaw} = \tau_{jaw} / r_{eff};\quad \tau_{jaw} = N \cdot \tau_{in} \cdot \eta^2 / 2$$

$\tau_{jaw} = 2 \times 0.922\ \text{N·m} \times 0.95^2 / 2 = 0.83\ \text{N·m}$. $r_{eff} = 59, 48, 33$ mm at 10, 40, 70 mm gap, so F = 14.1, 17.2, 25.0 N.

SIMULATED

The host plans; each station executes, and stop never disables.

HOST

Planner

Plans the path and sends timed segments

timed segments

STATION 1

Own clock

feed-hold · lease

STATION 2

Own clock

feed-hold · lease

STATION 3

Own clock

feed-hold · lease

STATION 4

Own clock

feed-hold · lease

≈ 18 ms

stop spread across stations (simulated)

10 / 10

lossy motions completed at 2–5 % packet loss (sim)

Feed-hold in path time

All four stations ramp to zero along the same geometric path.

Lease

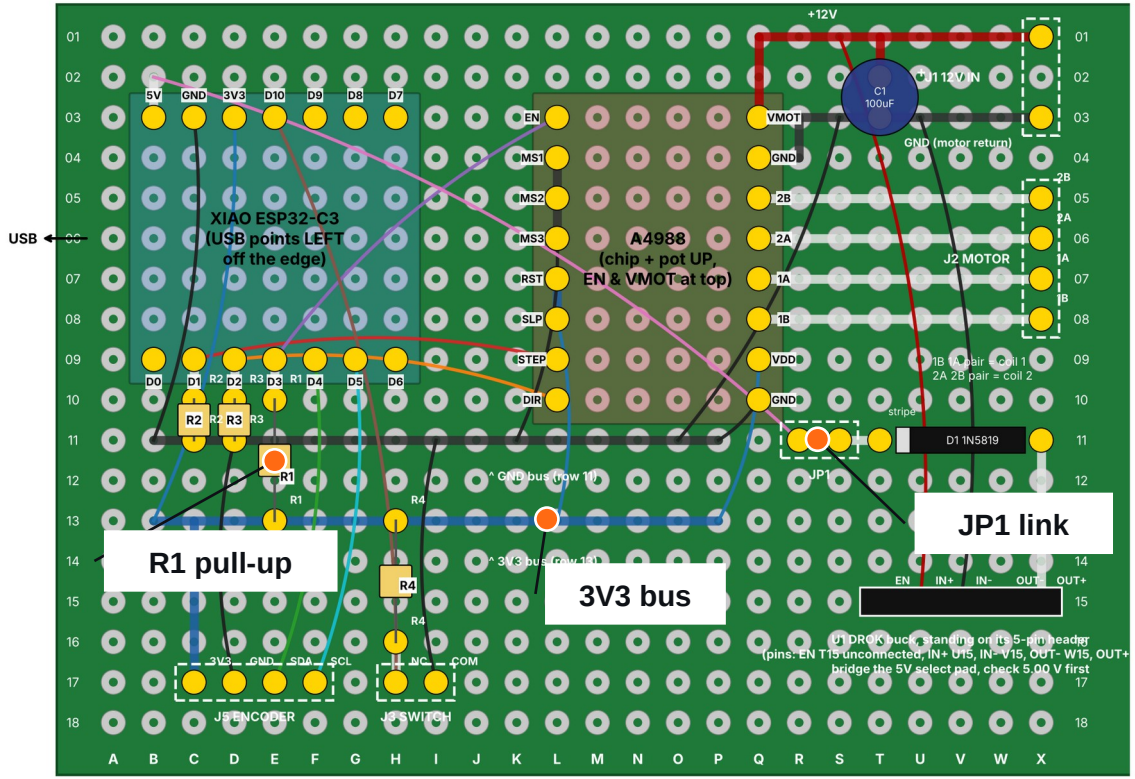
No valid host message in time: the station holds and tells its peers.

Stop never disables

Faults and pause hold torque. Only a confirmed disarm releases the load.

PLANNED

Rev D: soldered 3.3 V, driver boots disabled, no USB-buck fight.



COMPONENT SIDE VIEW. Thick grey/red/blue = bare wire or solder run on the bottom. Thin coloured = insulated wire on the bottom (crossings OK). Yellow = soldered hole.

Match module pin LABELS to the holes before soldering headers: XIAO 5V at B03, D0 at B09; A4988 EN at L03, VMOT at Q03, DIR at L10.

WHAT REV D CHANGED

3.3 V soldered, not jumpered.

10 k pull resistors: the driver boots disabled.

JP1 and diode D1: USB and the buck never fight.

Net-checked: every hole has one net, every net is connected.

PLANNED — not yet run

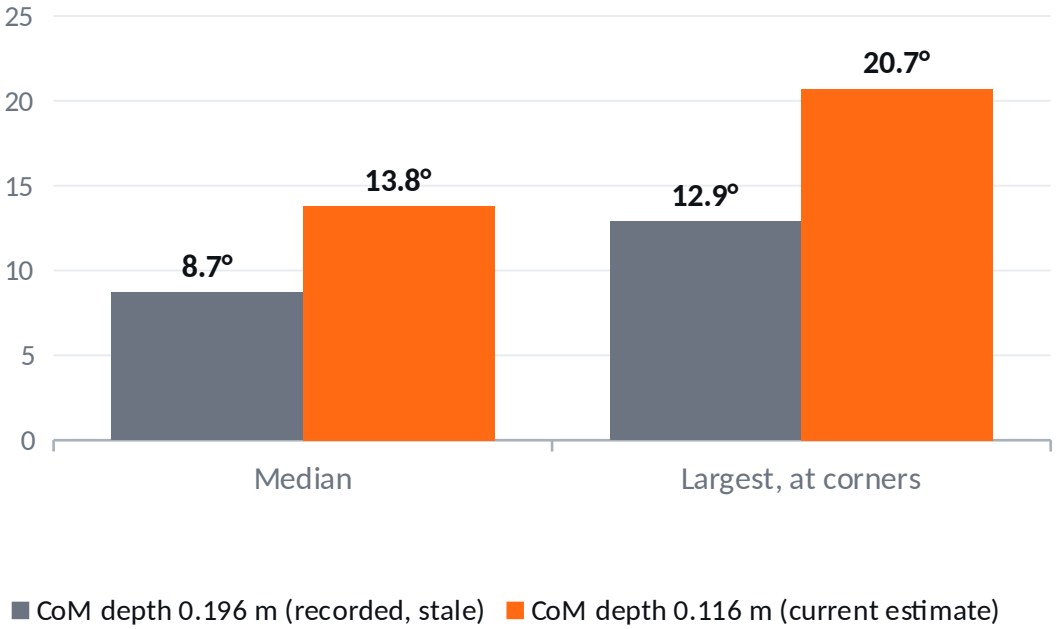
Meter checklist on the as-built board: VDD, SLEEP, ENABLE, buck output.

CALCULATED

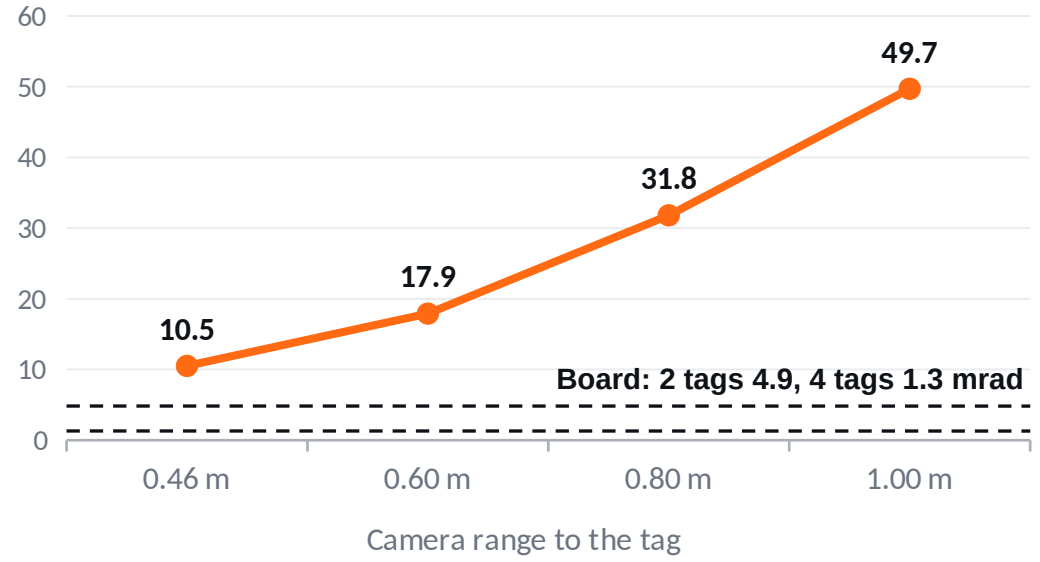
SIMULATED

Floor board cuts tilt noise up to 10×; hanging tilt still 13–21°.

HANGING TILT AT GRID POINTS (°)



TILT NOISE, ONE TAG, VS RANGE (mrad)



FUNDAMENTALS

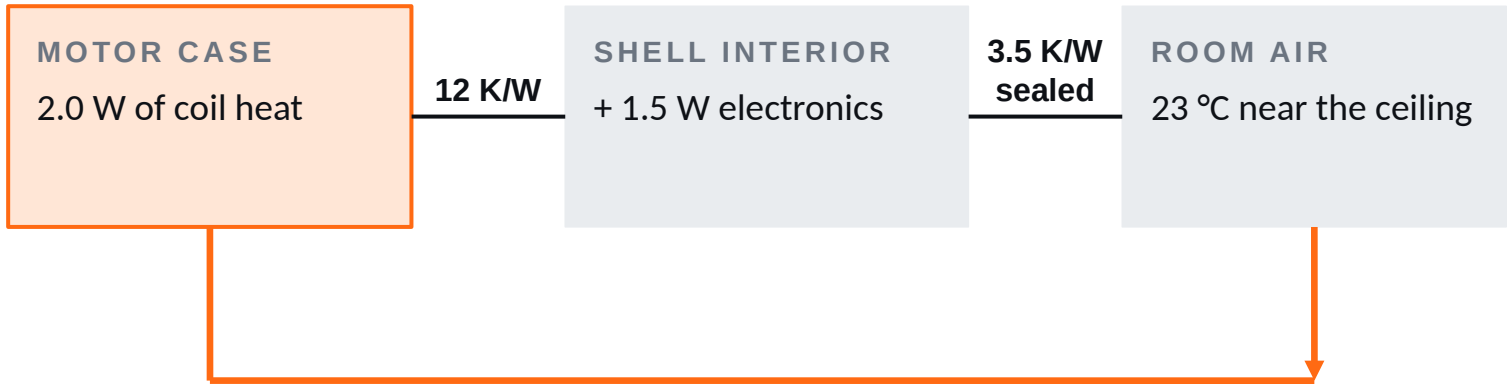
$$\theta \approx M_v / (W \cdot L); \quad \delta x = \sigma_\theta \cdot L$$

Corner tilt grows 1.61× when L falls from 0.196 to 0.116 m; 83 % then 92 % of grid points exceed 5°. Four tags in view: 0.68 mm vs 6.6 mm for one tag.

CALCULATED

PLANNED

The fin is a parallel heat path; PETG alone can't carry it.



Aluminium fin: 22.1 K/W as built, 10.9 K/W as drawn

Sealed shell, 1.06 A: 64 °C with no fin, 46 °C with the as-built fin, 40 °C with the drawn 1/8 in fin.

PETG VS ALUMINIUM

	Aluminium	PETG
k, W/m·K	160	0.2
Flange path, K/W	0.011	14.0
Spread, mm	74	2.6

PLANNED — not yet run

Fin FEA after the SolidWorks study; the hold test replaces the lumped model.

FUNDAMENTALS

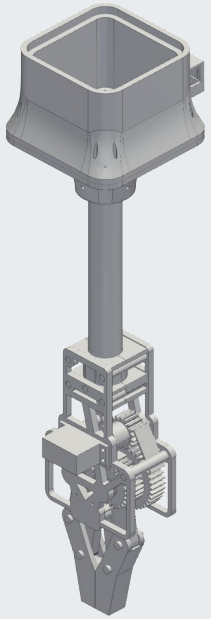
$$P_m = (T_m - T_{room}) / R_{fin} + (T_m - T_{shell}) / R_{ma}$$

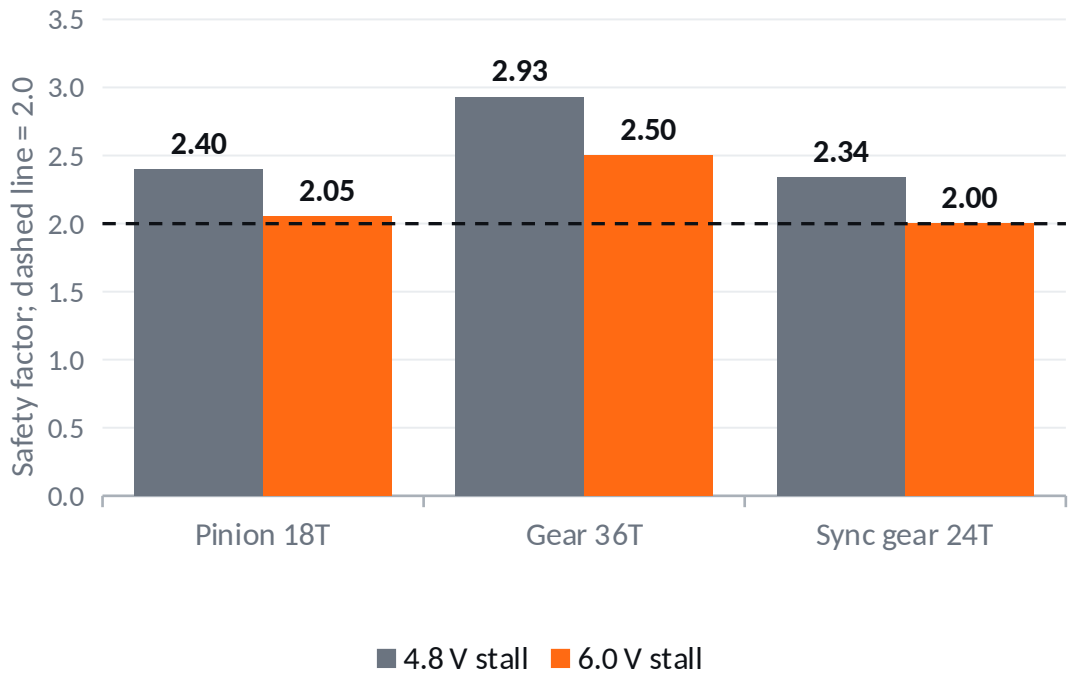
Equal geometry: $\sqrt{(160 / 0.2)} = 28\times$; spreading lengths are for a 0.48 mm sheet. Flange path, 5 mm PETG vs 3.2 mm aluminium: 1260× = 800× from k × 1.6× from thickness.

CALCULATED

SIMULATED

Open-frame v7 gearbox holds SF of about 2 or better at stall.





VERIFIED (REV 2)

Frame 143.5 g (limit 150 g).

Printed total 333 g; head 481 g.

72 checks pass; no interference over the sweep.

Open: servo ear pitch 47.0 vs 48.3 mm.

FUNDAMENTALS

$$SF = \sigma_{allow} / \sigma; \quad \sigma = F_t / (b \cdot m \cdot Y)$$

σ_{allow} = 22.5 MPa. At 6 V stall F_t = 79.9 N: pinion 11.0 MPa, SF 2.05. Sync gear, one jaw takes all: F_t = 113.9 N, SF 1.997.

Safety: one hard stop, anchors in studs, nobody underneath.



One hard stop

The switched power strip is the only stop. Faults and pause hold torque.



Stud anchors

Designed to screw into wall studs; screw pattern vs stud width and slide-in retention are still open. Line load is mostly pull-out and prying, plus some shear.



Nobody under it

Nobody stands under the robot while it moves.



Balance charging

Charge the 2S LiPo with a matched balance charger, pack off the robot.

No safety-rated function is claimed (ISO 13849 and IEC 61508 are not claimed).